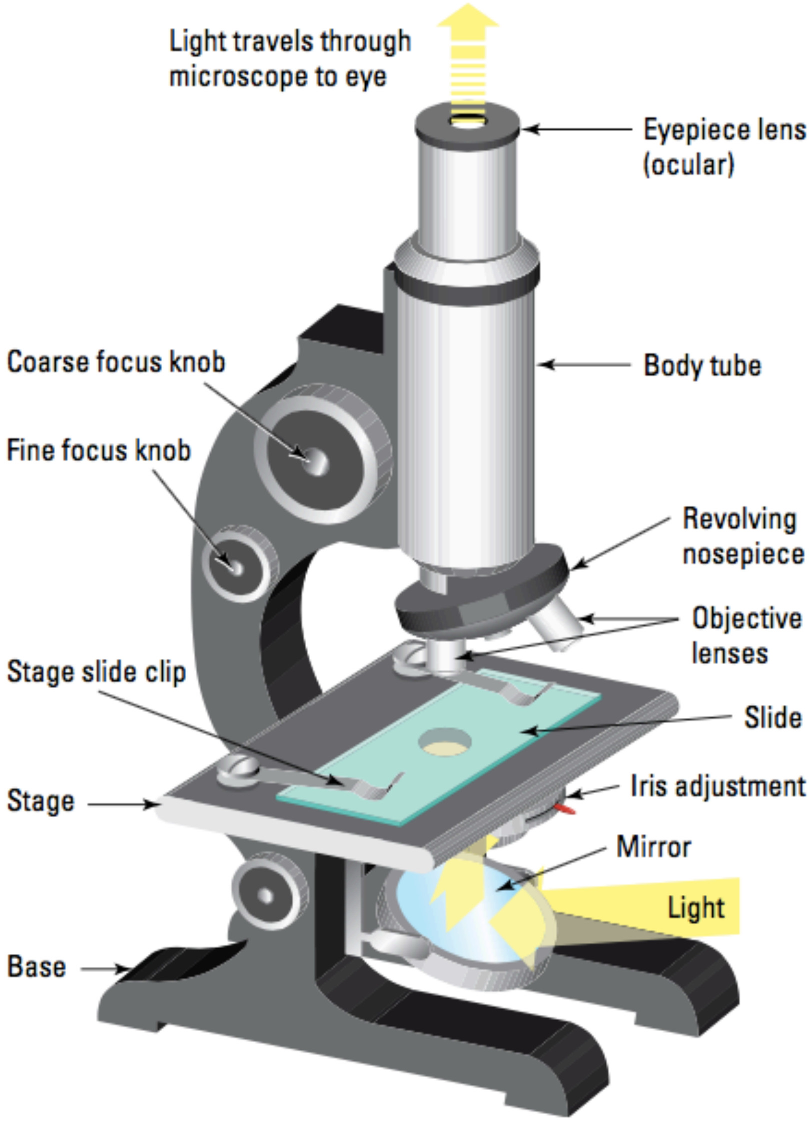
MICROSCOPES

The following diagram of the common microscope and its labelled parts need to be remembered.



Important Rules For Using Microscopes

- Make sure the mirror reflects light up through the stage.
- Focus on ***low power first***.
- ***Look from the side*** to adjust the objective lens ***just above the slide***.
- Use the ***coarse focus*** to wind the eye piece upwards under ***low power***.
- ***Look from the side when switching to higher power*** lenses to avoid hitting the slide.
- Use the ***fine focus*** when looking under ***high power***.

Activity

- Do Activity 3.1 p. 42 Science Quest 8.

Magnification

The two lenses that determine the **magnification** of your microscope are the eyepiece lens and the objective lens. Each lens has a number on it that signifies its magnification. Multiplying the eyepiece number by the objective lens number will give you the magnification of the microscope. For example:

- eyepiece: ×10
- objective: ×40
- magnification = ×400.

Field of view 4 mm (4000 μm) Magnification x40

Field of view 1.6 mm (1600 μm) Magnification x100

Field of view 0.4 mm (400 μm) Magnification x400

As the field of view gets smaller, the magnification gets larger.